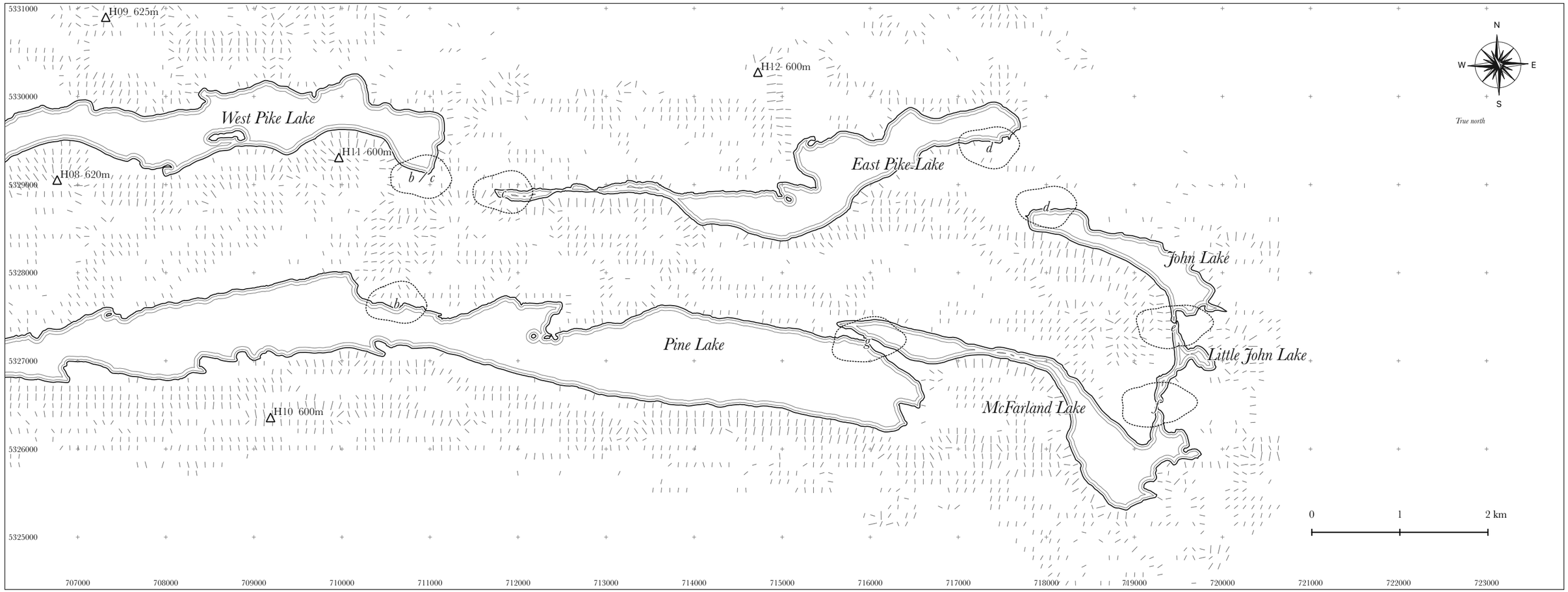
A DRAUGHT OF

Clearwater & the Eastern Lakes

The Pike lakes & eastern passage / A study in compass, distance & observation

THE EXERCISE / 03

Canoe country, Minnesota
A modern chart in the manner of an old draught



I. THE TWO CIRCUITS

A Clearwater - West Pike - Pine
Little Caribou - Caribou - Clearwater
B Clearwater - West Pike - East Pike - John
Little John - McFarland - Pine
Little Caribou - Caribou - Clearwater

Trace your course in pencil; choose the carries from the clues.

II. BY COMPASS & RULE

Grid crosses are 1 km apart. Map is grid-north up.
At chart centre: true bearing = grid bearing + 2.1°.
Magnetic bearing = grid bearing + 5.0°.
Variation 2.9° W; WMM2025, 12 October 2026.
Measure two landmark bearings; pencil their back-bearings.

Print at 100%. The line below must measure 50 mm.

III. SIGNS & OBSERVATIONS

Passage clue area
△ High ground (metres)
◆ Departure / return

Dotted areas are broad clues, not exact landing positions.
H points and hachures show bare-earth high ground.
Tree cover and landmark recognition are unverified.

EXPERIMENTAL TEACHING CHART | Exact carries deliberately withheld from the student edition.

MN DNR waters / USFS trails 2026-10-12 / MnGeo lidar 8 m / BGS WMM2025
NAD83 / UTM 15N (EPSG:26915); source notes & evidence: README and docs/